

RAG Learning Test Document

Introduction

Retrieval-Augmented Generation (RAG) combines information retrieval with large language models. Instead of relying only on the model's training data, RAG retrieves relevant documents and uses them as context.

What is an Embedding?

An embedding is a numerical representation of text. Similar pieces of text have vectors that are close together in vector space, enabling semantic search.

Chunking

Chunking divides a document into smaller pieces. Good chunk sizes improve retrieval accuracy and reduce irrelevant context.

Vector Database

A vector database stores embeddings and performs similarity search. ChromaDB is a lightweight vector database commonly used for learning RAG.

Prompt Engineering

A prompt builder combines the user's question with retrieved context before sending it to the language model.

Advantages

RAG reduces hallucinations, provides source-grounded answers, supports private documents, and keeps knowledge up to date without retraining the model.

Example Questions

What is an embedding? Why is chunking important? What are the benefits of RAG? What is ChromaDB used for? How does semantic search work?